25901657 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCECT101 - DATA COMMUNICATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Select the type of distortion occurs when different frequency components of a signal experience varying levels of attenuation [U] / [CO1]

   a). Amplitude distortion b). Phase distortion

   c). Delay distortion d). Frequency distortion

2. Specify the primary reason for converting analog signals into digital signals? [K] / [CO1]

   a). Digital signals are always faster

   b). Digital signals are less expensive to transmit

   c). Digital signals are less affected by noise and can be processed by computers

   d). Analog signals cannot be transmitted over long distances

3. Define the term 'bitrate'. [U] / [CO1]

a). The number of bits transmitted per unit time

b). The range of frequencies a signal occupies

c). The maximum frequency of a signal

d). The minimum frequency of a signal

4. Name the dielectric material used in coaxial cables [K] / [CO2]

a). Copper

b). Aluminum

c). PVC

d). Plastic or foam

5. Relate Infrared communication requires a direct line-of-sight because: [U] / [CO2]

a). Signals cannot pass through solid objects

b). Atmospheric conditions interfere with transmission

c). The signal strength decreases over long distances

d). Infrared waves scatter unpredictably

6. Label the main advantage of using serial ports over parallel ports. [U] / [CO3]

a). Faster data transmission over long distances

b). Simpler wiring and less signal interference

c). Better for high-speed data transfers

d). Supports multiple peripherals simultaneously

7. Which characteristics of an analog signal are changed to represent the lowpass analog signal in frequency modulation? [U] / [CO3]

a). Amplitude of the carrier signal

b). Frequency of the carrier signal

c). Phase of the carrier signal

d). Wavelength of the carrier signal

8. Select what is the main purpose of a channel encoder in a digital communication system? [U] / [CO3]

a). Compress data

b). Introduce redundancy for error detection and correction

c). Modulate the signal

d). Reduce bandwidth usage

9. In ASK transmission bandwidth is equal to . [U] / [CO3]

a). base bandwidth

b). twice base bandwidth

c). half the base bandwidth

d). four times the base bandwidth

10. The relationship between bit rate and baud rate is [U] / [CO3]

a). Bit rate is always equal to baud rate

b). Bit rate is always less than baud rate

c). Baud rate represents the number of signal units per second, while bit rate represents the number of bits per second

d). Bit rate and baud rate are unrelated

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Discuss about periodic and non-periodic signals. Provide two examples for each. [U] / [CO1]

12. Define and explain the three types of data flow: Simplex, Half-Duplex, and Full-Duplex, with suitable examples. [U] / [CO1]

13. Recall any five common applications of fiber optic communication. [K] / [CO2]

14. Compare FDM and WDM in terms of their applications. [U] / [CO3]

15. List the key benefits of using spread spectrum technology. [K] / [CO3]

16. How does data framing differ between synchronous transmission and asynchronous transmission. [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the factors that limit data rate in communication systems. Derive the Nyquist Bit Rate formula and Shannon Capacity formula, and compare their differences. [U] / [CO1]

18. Describe the construction of a twisted pair cable,including the purpose of twisted the wires and the different types of shielding(UTP and STP). [U] / [CO2]

19. Define DSSS and explain how it achieves bandwidth spreading. [U] / [CO3]

[10x2 = 20]